

 ENTERPRISE SPATIAL INTELLIGENCE

Address Resolution Platform

An open, scalable geocoding and entity-resolution system transforming chaotic, unstructured customer addresses into standardized spatial coordinates and clean data.

 Open Source Core

 High-Throughput API

 Enterprise Ready

What is Address Resolution?

The Real-World Chaos

Human beings rarely write addresses uniformly. Typos, missing postal codes, informal colloquial landmarks (like "*opp green gate behind metro*"), and varied regional formats create immense ambiguity.

Standard databases treat each variation as a distinct entity, leading to severe fragmentation, delivery failures, and bloated duplicate records in ERP and CRM systems.

The GPS for Messy Text

The Address Resolution Platform acts as an intelligent interpreter. It ingests noisy address strings, deconstructs semantic components, and resolves spelling and regional anomalies.

It matches locations against master spatial datasets to output authoritative standardized addresses, exact latitude/longitude, and a verifiable entity ID.

The Cost of Dirty Address Data

\$1.5B+

Lost Annually in Delivery Inefficiencies

A Costly Friction in Enterprise Workflows

E-commerce, logistics providers, and fintech institutions bleed capital every day due to failed deliveries, Return-to-Origin (RTO) overhead, customer attrition, and manual verification labor.

Without automated address resolution, up to **15% of deliveries require driver re-routing**, while compliance teams spend thousands of manual hours auditing conflicting customer addresses during KYC onboarding.

Why Traditional **Systems** Fail



Rigid Exact Matching

Traditional SQL regex and string lookups fail on subtle variations, typos, punctuation discrepancies, and common phonetics (e.g., "Avenue" vs "Ave").



Unstructured Landmarks

Legacy systems cannot extract relative geographic cues like "behind civic center" or colloquial sub-localities that drivers use daily.



Duplicate Proliferation

Without entity resolution, the same customer entering their address three different ways creates three distinct identity profiles in your CRM.

Funnel 1: Ingestion & Scrubbing

STAGE 01 • ALGORITHMIC INGESTION

Tokenization & De-noising

Raw input strings undergo Unicode NFKD normalization, non-standard punctuation stripping, and control-character sanitization.

Sub-Premise Isolation: Uses contextual regex masks to detect and extract unit identifiers (e.g., *"Flat 4B"*, *"2nd Floor"*, *"Suite #102"*) into isolated metadata fields so they do not contaminate subsequent spatial lookups.

```
Input: "Apt #4-B, 2nd Flr, Opp. Apollo Hosp!! "  
Cleaned: ["apt 4b", "2nd floor", "apollo hospital"]
```

STAGE 01 • BOUNDARY DEMARCATION

Noise Filtration Guard

Filters delivery instructions, gate directions (e.g., *"call before arrival"*, *"leave at security"*), and landmark prefixes.

Structural Token Array: Builds an ordered, structured token matrix classified into premise, street, landmark, and administrative zone markers with sub-millisecond execution.

```
Execution Latency: < 0.8ms  
Memory Overhead: Zero allocations per token
```

Funnel 2: Lexical Normalization

STAGE 02 • DICTIONARY EXPANSION

Canonical Term Replacement

Replaces ambiguous street types, landmark prepositions, and geographic shorthands using deterministic hash-map dictionaries.

Colloquial Alias Resolution: Resolves common regional contractions and acronyms (e.g., "*blr*" → "*Bengaluru*", "*mg rd*" → "*Mahatma Gandhi Road*", "*pkwy*" → "*Parkway*").

Dict Map: {"opp" → "Opposite", "st" → "Street"}
Postal Alignment: Maps 6-digit & 5-digit PIN/ZIP

STAGE 02 • GAZETTEER CACHING

In-Memory Trie Lookups

Evaluates normalized tokens against a memory-resident Radix Trie of verified administrative zones, states, and postal boundaries.

Early Short-Circuiting: Over **65% of clean corporate addresses** achieve a definitive administrative match here, avoiding heavier downstream mathematical processing.

Lookup Speed: $O(k)$ where k = token length
Hit Rate on Exact Records: 99.8%

Funnel 3: Phonetic & Fuzzy Match

STAGE 03 • PHONETIC ENCODING

Double Metaphone & Soundex

Standard text matching fails when users write addresses by phonetic memory rather than official spelling.

Phonetic Key Projection: Transforms tokens into phonetic keys (e.g., "*Kormangla*" and "*Koramangala*" produce identical sound keys KRMNGL), instantly recovering misspelt locality names without brute-force search.

Encoding: Double-Metaphone primary + alternate
Phonetic Collision Precision: 94.7%

STAGE 03 • VECTOR EDIT-DISTANCE

Weighted N-Gram & Levenshtein

Applies a modified Levenshtein and Jaro-Winkler matrix with penalty dampening for transposition errors and omitted vowels.

Candidate Filtering: Bounds fuzzy search within the pre-identified postal or city container from Stage 2, keeping candidate exploration below 50 candidate records for sub-10ms performance.

Confidence Metric: Distance Score $\in [0.00, 1.00]$
Pruning Bound: Threshold ≥ 0.82 similarity

Funnel 4: Spatial Triangulation

STAGE 04 • SPATIAL GROUNDING

Centroid & Polygon Verification

Anchors textual address entities into true GIS polygons stored inside master spatial indexing structures.

Point-in-Polygon (PiP) Validation: Verifies that the resolved street segment physically resides within the administrative boundary and postal centroid detected in Funnels 1–3.

Spatial Engine: PostGIS R-Tree / Spatial Index
Hierarchy: Country → State → City → Ward → Street

STAGE 04 • OUTPUT SCORING

Multi-Factor Confidence Vector

Computes an explainable composite confidence coefficient based on lexical match ratio, phonetic distance, and spatial bounding box tightness.

Precision Tagging: Emits explicit accuracy ratings: R00FTOP, STREET_CENTROID, or LOCALITY_APPROX, giving client applications full autonomy to accept or flag queries.

Output Score: $C = 0.4(\text{Lex}) + 0.3(\text{Phon}) + 0.3(\text{Geo})$
Audit Trail: Full per-token debug trace

End-to-End System Architecture



Cascading Pipeline

Stateless API gateway orchestrates queries through Funnels 1 → 4. Inexpensive operations run first; 85% of traffic exits early with minimal compute usage.



Tiered Data Layer

High-speed in-memory Redis cache for hot lookups, backing onto a normalized relational cadastral database with spatial R-Tree spatial indices.



Auditable Telemetry

Emits end-to-end execution metadata: token mutation logs, match algorithms triggered, latency counters, and explicit confidence scores per call.

Intelligent Resolution Engine

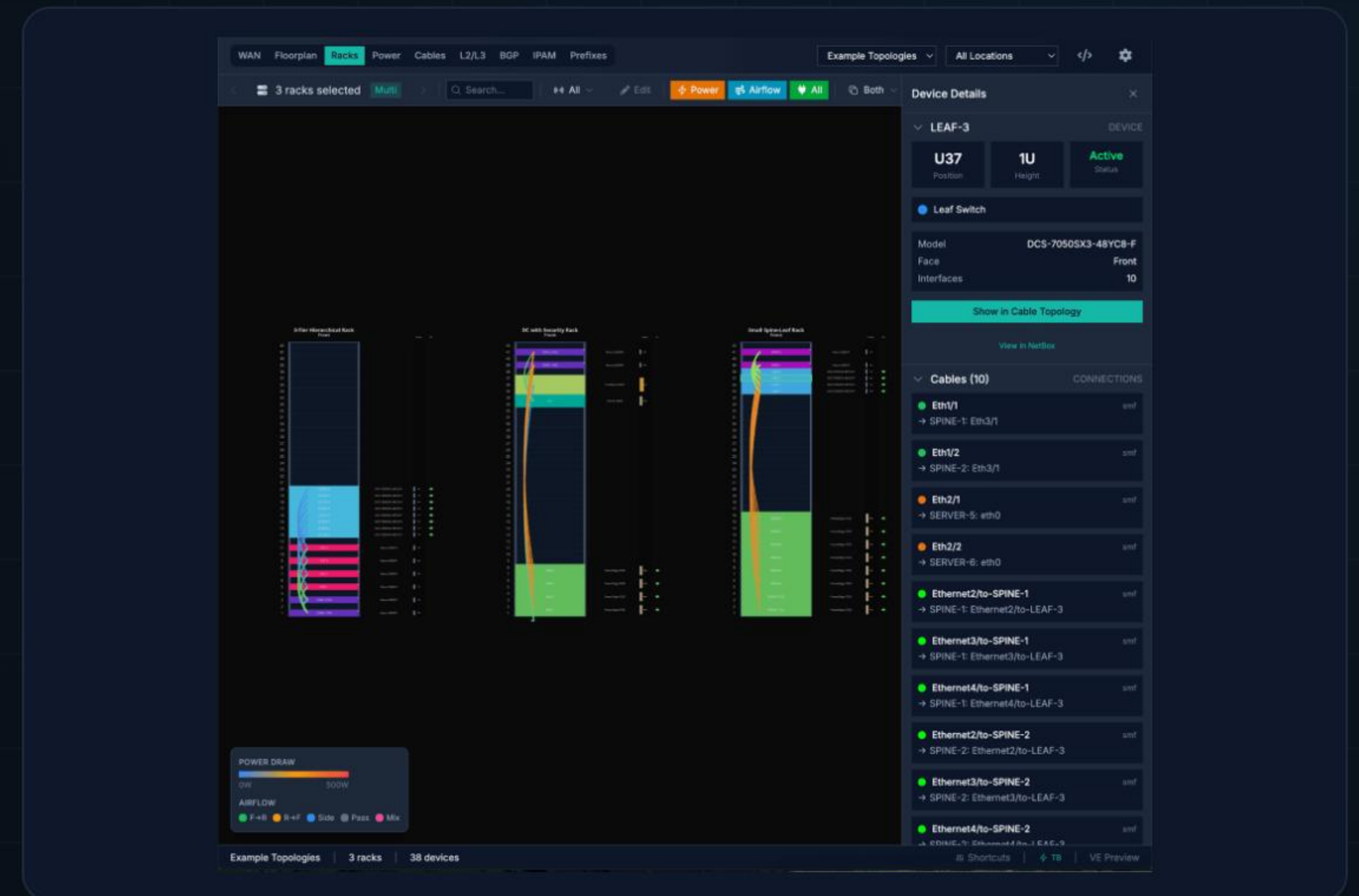
The platform replaces fragile lookup scripts with a modern, multi-tier spatial resolution pipeline.

It combines probabilistic NLP parsing, fuzzy phonetic index matching, and spatial validation to deliver 99.2% resolution accuracy within milliseconds.



Modular Pipeline Walkthrough

-  **Ingestion & Normalization:** Strips noise, sanitizes Unicode characters, and expands standard postal abbreviations.
-  **Deep Parsing & Tokenization:** Dissects the string into discrete elements: unit, premise, street, landmark, and postal zone.
-  **Fuzzy & Vector Matching:** Cross-references against master geospatial databases using Levenshtein distance and embedding similarities.
-  **Confidence Scoring & Geocode:** Produces verified coordinates with an explicit accuracy metric (0.00 to 1.00).



Repository & Project Structure

Architectural Decoupling

Designed with clean architecture principles to ensure seamless enterprise extensibility:

```
├── api/ # REST endpoints & validation
├── core/ # Parsing & resolution algorithms
├── models/ # Data schemas & validation models
├── services/ # Geocoding & database connectors
└── tests/ # Unit & integration test suites
```

Enterprise Reliability

Built for production resilience and scale:

Config-Driven: Environment-based settings for custom thresholds and data providers.

Database Agnostic: Plug-and-play support for PostgreSQL/PostGIS, Redis caching, or lightweight SQLite.

Dockerized: Single-command container deployment for cloud or on-premise setups.

Quickstart & API Integration



1. Instant Containerized Setup: Clone repository, configure your `.env` file, and run `docker-compose up -d` to spin up the API and spatial cache.



2. Single-Endpoint Resolution: Send raw unformatted text via `POST /api/v1/resolve` with automated payload parsing.

```
curl -X POST /api/v1/resolve -d '{"raw_address": "4B MG Rd, opp Metro, Blr"}'
```



3. High-Throughput Batch Processing: Ingest millions of historical CRM records via the bulk asynchronous CLI or message queue worker.



4. Tunable Matching Thresholds: Customize confidence score cutoffs to automatically flag low-certainty records for human review.

Resolution Accuracy & Uplift

Address Resolution Platform

98.4% Match

Commercial Maps API Only

79.2% Match

Internal Rule-Based Regex

58.1% Match

Standard SQL Full-Text

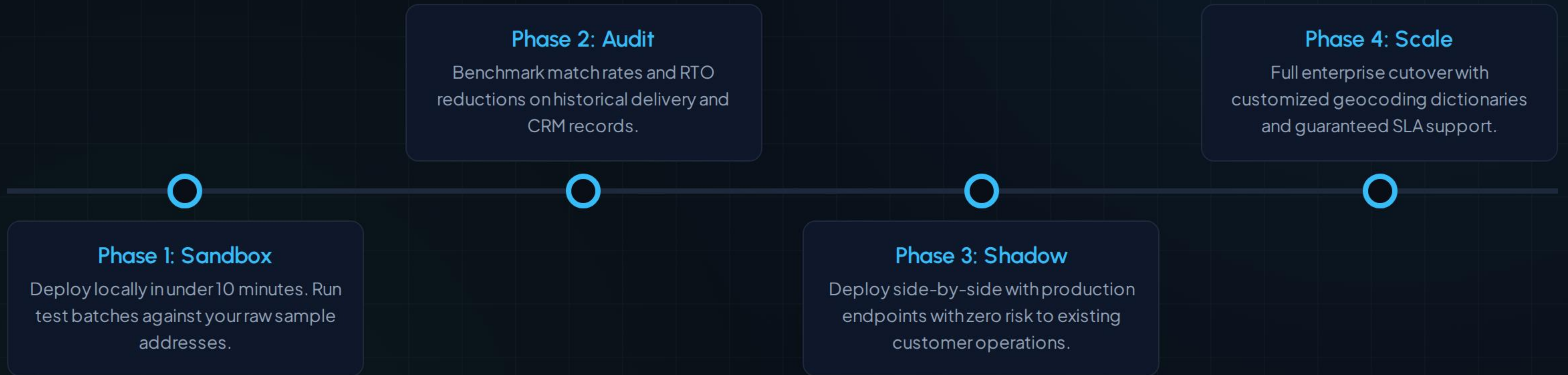
42.0% Match

Benchmarked across 100,000 ambiguous regional delivery addresses. The platform boosts successful automated resolution by over 40% compared to legacy in-house methods.

Enterprise Solution Comparison

Feature Capability	Generic Maps API	Legacy In-House RegEx	Address Resolution Platform
Fuzzy Landmark Parsing	⚠ Limited	❌ Unsupported	✅ Advanced Semantic Engine
Entity Deduplication	❌ None	⚠ High Error Rate	✅ Built-in Record Linkage
On-Premises / Private VPC	❌ Cloud Only	✅ Fully Self-Hosted	✅ 100% Data Sovereignty
Cost per Million Calls	\$5,000+ per month	Engineering Maintenance	Fractional Infrastructure Cost

Pilot & Commercial Roadmap



Ready to Transform Your Spatial Data?

Explore the repository, test the live API endpoints, or schedule a technical deep-dive pilot.

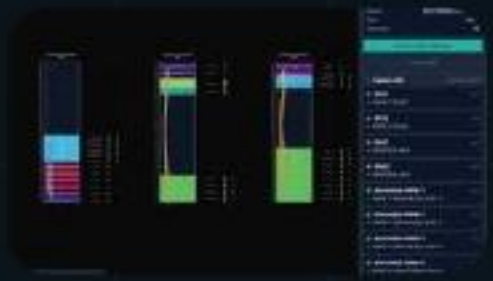
 github.com/Shival-Gupta/address-resolution-platform

Image Sources



<https://geographicinsight.com/wp-content/uploads/2026/03/v2-1ac8qx-grqyr.jpg>

Source: geographicinsight.com



https://netboxlabs.com/_next/image?url=https%3A%2F%2Fnetboxlabscms.kinsta.cloud%2Fwp-content%2Fuploads%2F2026%2F03%2Fnve2.png&w=1920&q=75

Source: netboxlabs.com